

IES FRANCISCO DE GOYA

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CURSO: DAM 2

Entornos de Desarrollo: Banking Kata

A continuación se presentarán evidencias sobre el ejercicio planteado en el aula virtual, se presentarán tanto las imágenes como el texto plano del fragmento del código:

Código del programa

```
package Banking;

import java.time.LocalDate;
import java.time.format.DateTimeFormatter;
import java.util.ArrayList;
import java.util.List;

public class Account {

    private static final DateTimeFormatter DATE_FORMATTER =
        DateTimeFormatter.ofPattern("d.M.yyyy");

    private int balance = 0;

    private final List<Transaction> transactions = new ArrayList<>();

    private LocalDate currentDate;

    public Account() {
        this.currentDate = LocalDate.now();
    }

    public void deposit(int amount) {
        balance += amount;

        transactions.add(
            new Transaction(currentDate, amount, balance)
        );
    }

    public void withdraw(int amount) {
        balance -= amount;

        transactions.add(
            new Transaction(currentDate, -amount, balance)
        );
    }
}
```

```
public void setDate(LocalDate date) {
    this.currentDate = date;
}

public String printStatement() {

    StringBuilder statement = new StringBuilder();

    statement.append("Date          Amount   Balance");

    for (int i = transactions.size() - 1; i >= 0; i--) {

        Transaction transaction = transactions.get(i);

        statement.append("\n");

        statement.append(
            String.format(
                "%-12s%+5d%9d",
                transaction.date().format(DATE_FORMATTER),
                transaction.amount(),
                transaction.balance()
            )
        );
    }

    return statement.toString();
}

private record Transaction(
    LocalDate date,
    int amount,
    int balance
) {
}
}
```

Pruebas

```
package BankingTest;

import org.junit.jupiter.api.Test;
import java.time.LocalDate;
import Banking.Account;

import static org.junit.jupiter.api.Assertions.assertEquals;

class AccountoTesto {

    @Test
```

```
void ImprimirExtractoDespuesDeUnDeposito() {
    Account account = new Account();

    account.setDate(LocalDate.of(2015, 12, 24));
    account.deposit(500);

    String statement = account.printStatement();

    assertEquals(
        "Date          Amount  Balance\n" +
        "24.12.2015    +500      500",
        statement
    );
}

@Test
void ImprimirExtractoDespuesDeDepositoYRetiro() {
    Account account = new Account();

    account.setDate(LocalDate.of(2015, 12, 24));
    account.deposit(500);

    account.setDate(LocalDate.of(2016, 8, 23));
    account.withdraw(100);

    String statement = account.printStatement();

    assertEquals(
        "Date          Amount  Balance\n" +
        "23.8.2016      -100      400\n" +
        "24.12.2015    +500      500",
        statement
    );
}

@Test
void ActualizarSaldoDespuesDeDeposito() {
    Account account = new Account();

    account.setDate(LocalDate.of(2015, 12, 24));
    account.deposit(500);
    account.deposit(300);

    String statement = account.printStatement();

    assertEquals(
        "Date          Amount  Balance\n" +
        "24.12.2015    +300      800\n" +
        "24.12.2015    +500      500",
        statement
    );
}

@Test
```

```

void ActualizarSaldoDespuesDeRetiro() {
    Account account = new Account();

    account.setDate(LocalDate.of(2015, 12, 24));
    account.deposit(1000);

    account.setDate(LocalDate.of(2016, 8, 23));
    account.withdraw(300);

    String statement = account.printStatement();

    assertEquals(
        "Date          Amount  Balance\n" +
        "23.8.2016        -300    700\n" +
        "24.12.2015      +1000    1000",
        statement
    );
}

@Test
void RetornarEncabezadoParaCuentaNueva() {
    Account account = new Account();

    assertEquals(
        "Date          Amount  Balance",
        account.printStatement()
    );
}
}

```

Capturas de pantalla del funcionamiento

